

		Logistic Regression (Best Baseline)	Naive Bayes (Best Baseline)	Linear SVC (Best Classical)	Random Forest	Embedding + BiLSTM	Embedding + Fine Tuning	Ensembling(Embedding+Fine tuning+LR+Linear SVC)
								Input file is already lowercase. Missing values filled with ". No extra regex cleaning.
Preprocessing	Clean Text	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Stop Words	No	No	No	No	No	No	No
	Lemmatization	No	No	No	No	No	No	No
	Stemming	No	No	No	No	No	No	No
Vectorizer / Tokenizer	Main representation	BoW raw	BoW raw	TF-IDF	BoW	DistilBERT tokenizer	DistilBERT tokenizer	DistilBERT tokenization + TF-IDF word/character features
	N-gram range	(1,1)	(1,1)	(1,2)	(1,1)	Not applicable	Not applicable	Word TF-IDF: (1,3); Character TF-IDF: (3,6)
	Combined features	No	no	TF-IDF_raw + TF-IDF_stemm	No	No	No	hstack(word TF-IDF + char_wb TF-IDF) + soft-voting with DistilBERT models
Trainer Params	Max Features	5000	5000	15000	5000	num_words=10000	max_length=128	200,000 word + 200,000 char; actual combined features: 311,744
	min_df	Default	Default	3	Default	Not applicable	Not applicable	Word TF-IDF: 1; Character TF-IDF: 2
	max_df	Default	Default	0.85	Default	Not applicable	Not applicable	Default 1.0
	Model / architecture	Logistic Regression	MultinomialNB	Linear SVC	RandomForestClassifier	BiLSTM(64)	distilbert-base-uncased	Soft-voting ensemble: DistilBERT seed 42 + DistilBERT seed 0 + Calibrated LinearSVC + Logistic Regression
	Trainer / optimizer	sklearn fit	sklearn fit	sklearn fit	sklearn fit	Adam + EarlyStopping	HuggingFace Trainer	Hugging Face Trainer; AdamW default optimizer; learning rate 2e-5; batch size 16; weight decay 0.01
	Epochs	Not applicable	Not applicable	Not applicable	Not applicable	Up to 10 (early stopping)	2	2
Performance	Training accuracy	0.972	0.9451	0.9987	0.9297	0.991728	0.9961	0.9961
	Training loss	0.0989	0.1593	0.071	Not recorded	0.02436	0.01685	Seed 42: 0.0781; Seed 0: 0.0789
	Validation accuracy	0.9477	0.9409	0.9537	0.9166	0.966184	0.98682	98.79
	Validation loss	0.1406	0.1826	0.1675	Not applicable	0.121377	0.05787	Not recorded
	Precision	Not recorded	Not recorded	Not recorded	Not recorded	0.96	0.99	0.99
	Recall	Not recorded	Not recorded	Not recorded	Not recorded	0.96	0.99	0.99
	F1-score	Not recorded	Not recorded	Not recorded	Not recorded	0.96	0.99	0.99
	Confusion matrix	Not recorded	Not recorded	Not recorded	Not recorded	[[3465, 64], [184, 3118]]	[[3473, 42], [48, 3268]]	[[3466 49] [37 3279]]
Comments	Additional comments	Best simple baseline	Good baseline but lower than LR/SVC	Best classical model; strong feature engineering	Removed from later experiments; weaker and slower	Strong neural embedding model	Best model without ENSEMBLING	Best overall final model